

6 (a) The DNA in the nucleus is known as nuclear DNA.

- (i) In the cells of the grasshopper, *Chorthippus brunneus*, 20% of the nucleotides in nuclear DNA contain thymine.

Calculate the percentage of nucleotides in the nuclear DNA of *C. brunneus* that contain guanine **and** explain your answer in terms of the structure of DNA.

percentage

explanation

.....

.....

.....

..... [3]

- (ii) State another location, **other than** the nucleus, where DNA occurs in cells of *C. brunneus*.

..... [1]

(b) Fig. 6.1 is a diagram of a molecule of tRNA.

The region labelled **R** shows detail of part of the tRNA molecule.

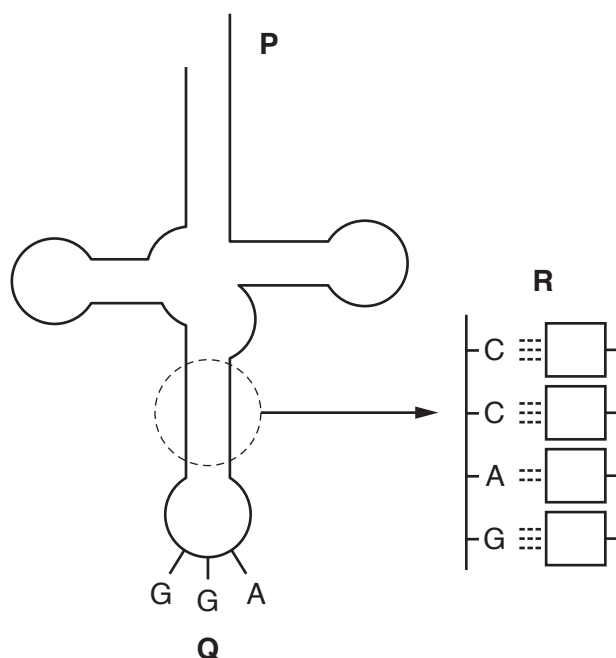


Fig. 6.1

(i) Complete Fig. 6.1 by writing the sequence of bases in the region labelled **R**. [1]

(ii) State the name of region **Q** **and** explain the role of region **Q** in translation.

name

explanation

.....

.....

..... [3]

(iii) State the function of region **P**.

.....

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..... [1]

[Total: 9]